

Experiment 1: Architecture Comparison

Proving Analytic Force Direction Is Necessary for Long-Horizon Orbital Stability

Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

SIMON V3 - Paper-Ready Section - May 2026

1. Scientific Question and Contribution

Central question: Does constraining force direction to analytic geometry provide long-term orbital stability benefits that an equally accurate learned-direction model cannot replicate, even after training to sub-degree directional accuracy?

The V2 paper positioned SIMON as a physically constrained force model, with force direction always computed from exact geometry. However, V2 never tested whether this constraint is doing useful work: would a network that learns force direction, rather than fixing it analytically, perform equivalently? Without a controlled comparison, the contribution of the direction constraint is an architectural claim, not an experimental finding.

This experiment provides that controlled comparison. A Vector NN baseline is trained to predict force direction with sub-degree accuracy, while all other conditions are held identical to SIMON. The result establishes whether analytic direction enforcement is necessary or merely an architectural preference.

This experiment is one of two pillars supporting V3's central principle: long-horizon stability in chaotic three-body neural integration requires two complementary physical constraints. This experiment isolates the force direction constraint specifically. The adaptive sub-stepping constraint is isolated in Experiment 4.

2. Experimental Design

The design isolates a single variable: whether force direction is computed analytically or predicted by the neural network. All other conditions are identical across models.

Table 1: Three-Model Comparison Design

The three models differ only in what the neural network predicts and where force direction comes from.

Model	NN predicts	Direction source	Magnitude source
ias15	Nothing (ground truth)	Exact - 15th-order Gauss-Radau	Exact analytic
SIMON	Scalar correction factor c	Exact geometry - float64 (never predicted by NN)	$c \times G \cdot m_i \cdot m_j / r_{\text{soft}}^3$
Vector NN	Unit direction vector	NN prediction - float32 approx. 0.04 deg residual	Exact analytic: $G \cdot m_i \cdot m_j \cdot r / r_{\text{soft}}^3$

The Vector NN receives strictly more information than SIMON: it receives the geometric direction components $[r_x/r, r_y/r, r_z/r]$ as additional inputs. If the Vector NN had fewer inputs, a poor result could be attributed to information deficit. By giving it more information, any stability disadvantage must arise from the learned approximation of direction itself, not from missing inputs.

2.1 Vector NN Architecture and Training

The Vector NN uses the same hidden-layer structure as SIMON (3 hidden layers, 32 units, SiLU activations) but with 7 inputs and 3 outputs instead of 3 inputs and 1 output.

Table 2: Architecture Comparison - SIMON vs Vector NN

Component	SIMON	Vector NN
Input features	$[\log(r_{\text{soft}}), \log(m_i), \log(m_j)]$ - 3 scalars	$[r_x/r, r_y/r, r_z/r, \log(r_{\text{soft}}), \log(r), \log(m_i), \log(m_j)]$ - 7 features
Output	$\log(c)$ - scalar correction factor	$[dx, dy, dz]$ - unit direction vector
Parameters	2,273	2,467
Training epochs	5,000	20,000
Training time	9 seconds	33 seconds
Loss function	MSE on $\log(c)$	MSE on unit direction vectors (range [0,4])
Best val MSE	1.1×10^{-5}	4×10^{-8} (unit direction MSE)

Figure 1: Vector NN Training Curve

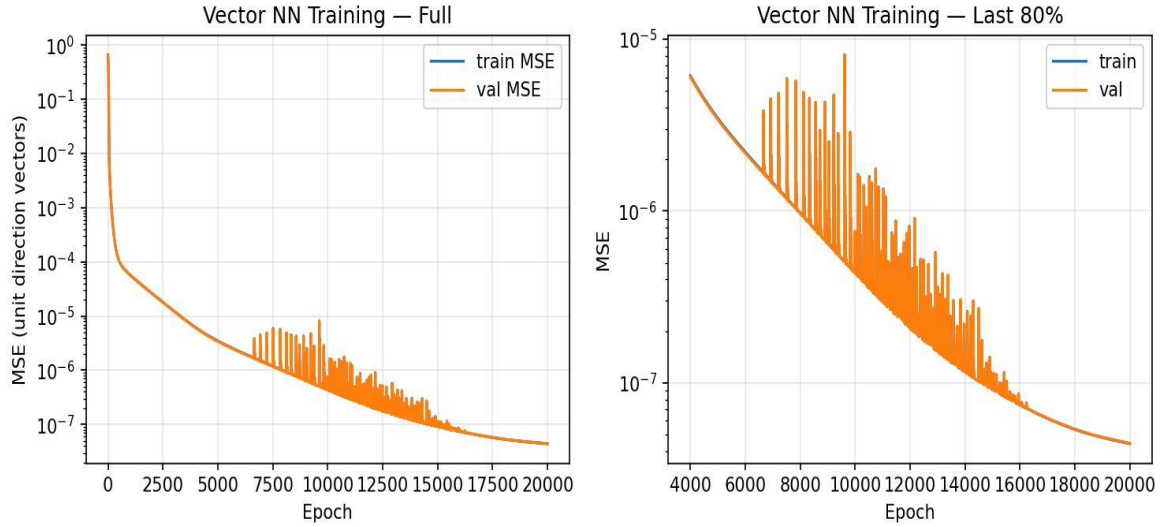


Figure 1. Vector NN training curve over 20,000 epochs. Left: full training run (log scale). Right: final 80% showing convergence. Training and validation MSE are indistinguishable, confirming no overfitting. Val MSE = 4×10^{-8} at epoch 20,000, corresponding to directional accuracy of approximately 0.04 degrees across the tested distance range.

Table 3: Vector NN Directional Accuracy Sanity Check

Post-training verification confirms sub-degree directional accuracy across the full operational distance range.

r (AU)	F _{true}	F _{infer}	Direction error (deg)	Magnitude error (%)
1.5×10^{-4}	3.98×10^6	3.98×10^6	0.040	0.000
1.0×10^{-3}	8.79×10^5	8.79×10^5	0.000	0.000
1.0×10^{-2}	9.99×10^3	9.99×10^3	0.000	0.000
1.0×10^{-1}	1.00×10^2	1.00×10^2	0.000	0.000
1.0	1.00	1.00	0.020	0.000
5.0	4.00×10^{-2}	4.00×10^{-2}	0.020	0.000

Table 3. Vector NN post-training sanity check ($m_i = m_j = 1$ solar mass, x-axis test pairs). Force magnitude error is analytically exact (0.000%) across all distances. Maximum directional error is 0.04 degrees. This confirms the Vector NN reached its accuracy ceiling and is a fair, well-trained baseline for comparison.

3. Results

Important note on result structure: Per mentor guidance, the gate-sensitivity analysis (Finding 2 below) is the primary scientific evidence and is presented first. The trajectory and divergence figures (Finding 1 and 3) serve as illustrations of the mechanistic finding, not as standalone claims.

Finding 1 (Primary): Gate Sensitivity Analysis - The Mechanistic Proof

To determine whether the Vector NN ejection arises from a single anomalous prediction or from the cumulative effect of many individually acceptable errors, the direction gate threshold was progressively tightened and the simulation re-run at each setting.

Table 4: Gate Sensitivity Analysis Results

Gate setting	Interpretation	Ejection at year 35?	Final RMS (AU)
$\cos < 0.866$ (30 deg gate)	Standard gate - triggers on large deviations only	YES - unchanged	83.0
$\cos < 0.9999985$ (0.1 deg gate)	Tighter than NN's own 0.04 deg accuracy - gate never fires on any prediction	YES - identical timing	83.0

Mechanistic interpretation: When the gate is set tighter than the NN's own demonstrated accuracy ($\cos < 0.9999985$, corresponding to 0.003 degrees), it never fires on any prediction. Despite this, ejection occurs with identical timing and identical final RMS of 83.0 AU. This proves the failure arises from the cumulative effect of many individually acceptable 0.04-degree errors, not from any single anomalous prediction that a gate could catch. **The calculation:** At $dt = 0.04$ yr over 100 years, there are approximately 2,500 integration steps. A directional error of 0.04 degrees per close-encounter step accumulates to approximately 100 degrees of non-radial impulse. Gravitational force is a central force by definition: it acts exactly along the line connecting two bodies. Non-radial force components have no physical basis, and in a chaotic system they compound into non-physical energy injection that eventually ejects a body regardless of how the gate is set. No gate-based quality filter can replicate the structural guarantee of zero perpendicular force that SIMON's analytic direction provides.

Finding 2 (Illustration): Quantitative Simulation Results

For the tested initial condition (masses 1.0, 0.01, 0.005 solar masses; 100-year horizon; $dt = 0.04$ yr), the three models produce the following outcomes. Note that this represents a single initial condition; the gate-sensitivity analysis in Finding 1 above provides the mechanistic evidence that the result is robust to the specific trajectory realised.

Table 5: Quantitative Results Summary

Metric	ias15 (ground truth)	SIMON	Vector NN
Divergence rate lambda (10-50 yr)	- (reference)	0.1655 /yr	0.1656 /yr
Final RMS error at T=100 yr	0.000 AU	1.54 AU	83.0 AU
All bodies bounded at T=100 yr?	YES	YES	NO - ejection approx year 35
Speedup vs ias15 (dt=0.04)	1.00x	1.06x	1.04x
NN invocation rate	-	0.45% of pair evaluations	1.01% of pair evaluations
Directional precision	float64 exact	float64 exact (analytic)	approx 0.04 deg residual (NN)

Table 5. Quantitative comparison for the tested initial condition. Both models show identical divergence rates during the bounded phase ($\lambda = 0.1655$ vs $0.1656/\text{yr}$), confirming both correctly reproduce the chaotic dynamics of the three-body system. The stability difference is qualitative, not quantitative: SIMON remains bounded; the Vector NN does not.

Finding 3 (Illustration): Divergence Rate and Trajectory Figures

The following figures illustrate the mechanistic finding from Finding 1. They show what the accumulated non-radial impulse looks like in terms of trajectory and divergence rate. Both models track identically during the bounded phase, confirming they reproduce the same underlying chaotic physics.

Figure 2: Divergence Rate - SIMON vs Vector NN vs ias15

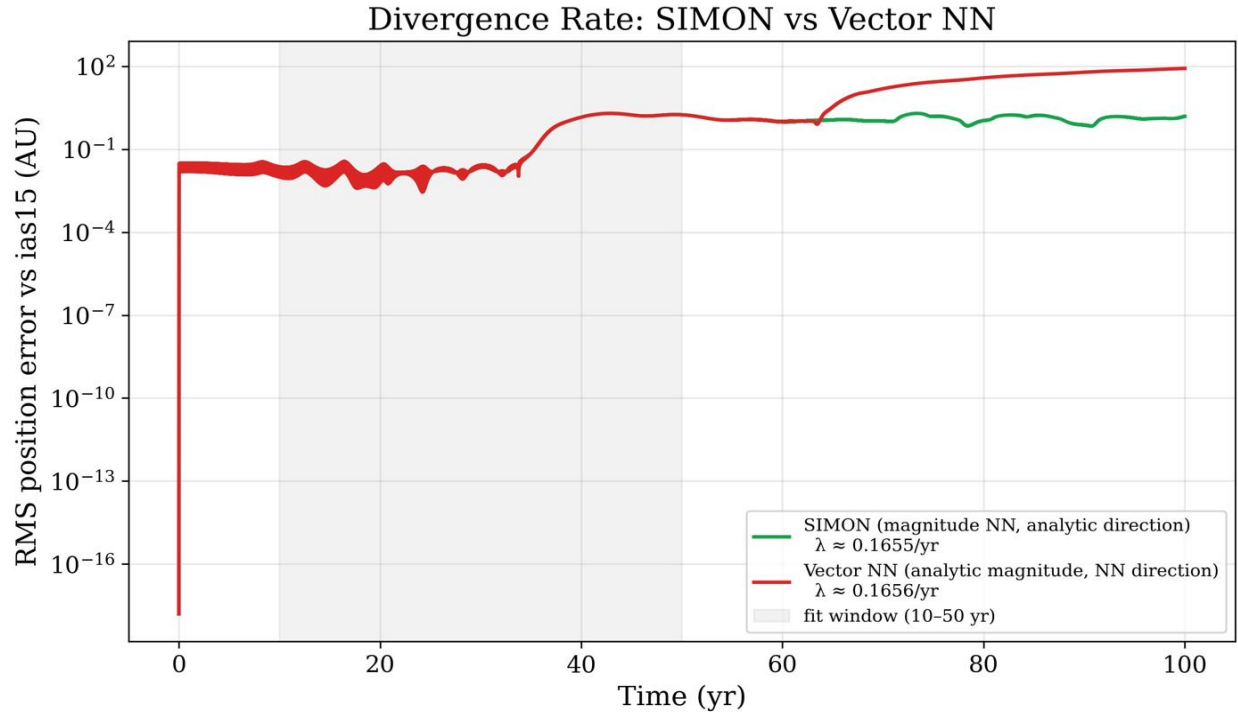


Figure 2. RMS position error vs ias15 over 100 years. SIMON (green dashed) maintains bounded orbital evolution with $\lambda = 0.1655/\text{yr}$ throughout the full simulation. Vector NN (red) tracks identically for approximately the first 35 years - demonstrating both models reproduce the same chaotic physics during the bounded phase - then undergoes body ejection, reaching 83 AU by year 100. The fit window (shaded, 10-50 yr) confirms λ values are indistinguishable during the bounded phase (0.1655 vs 0.1656/yr). The divergence in outcome is qualitative, not quantitative: bounded vs ejected.

Figure 3: XY Trajectory Overlay - All Three Bodies

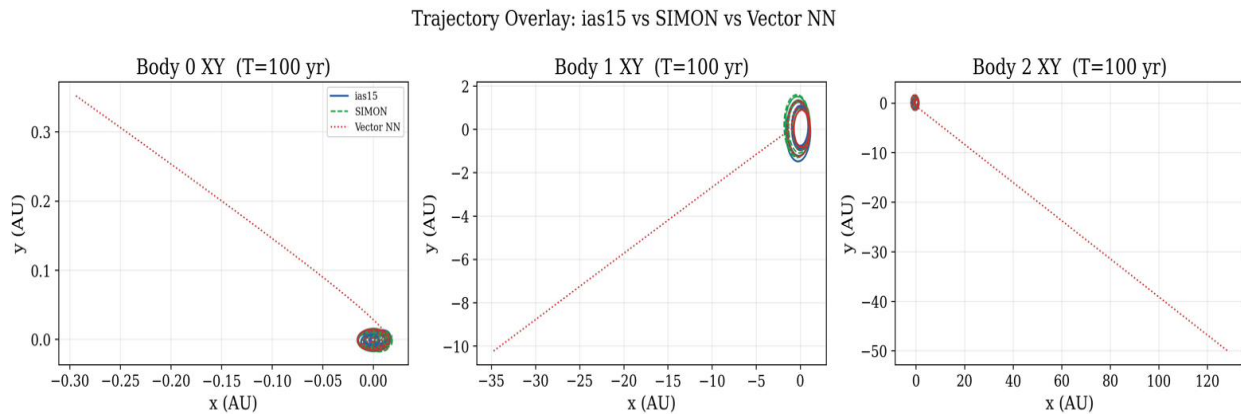


Figure 3. XY trajectory overlay for all three bodies over 100 years. ias15 (blue solid) and SIMON (green dashed) produce overlapping bounded orbits throughout. Vector NN (red dotted) follows the same trajectories until approximately year 35, after which accumulated non-radial impulses trigger ejection: Body 0 reaches 125+ AU and Body 1 reaches 35+ AU. This figure illustrates the physical consequence of the mechanism identified in Finding 1 (Table 4).

4. Connection to the Four-Run Ablation Framework

This experiment provides Run B in the four-run ablation framework that forms the spine of V3's Results section. Each run isolates one component of SIMON's architecture by removing it while holding everything else constant.

Table 6: Four-Run Ablation Framework

Run	Adaptive stepping	Force direction	NN correction	Result	What it proves
A (Experiment 4)	OFF	Analytic	ON	Ejection 161.7 AU	Adaptive stepping is necessary for stability
B (This experiment)	ON	Learned (Vector NN)	N/A	Ejection 83.0 AU	Analytic force direction is necessary for stability
C (Experiment 6)	ON	Analytic	OFF	Bounded 1.67 AU	NN correction provides accuracy (8.5% improvement), not stability
D (SIMON baseline)	ON	Analytic	ON	Bounded 1.54 AU	Both constraints active: best result

Table 6. Four-run ablation framework. Run B (this experiment) isolates the force direction constraint. Run A (Experiment 4) isolates adaptive stepping. Run C (Experiment 6) isolates the NN correction contribution. Run D is the full SIMON system. Each run removes exactly one component; the outcome isolates that component's role.

The role of the NN correction (Run C vs Run D) is addressed by the ablation study in Experiment 6. Removing the NN correction while retaining adaptive stepping and analytic direction yields bounded orbits with 8.5% larger final RMS error (1.67 vs 1.54 AU). This confirms the NN correction provides force accuracy at close encounters and is not the primary stability mechanism. The figure below shows this comparison.

Figure 4: NN Ablation Divergence - Run C vs Run D

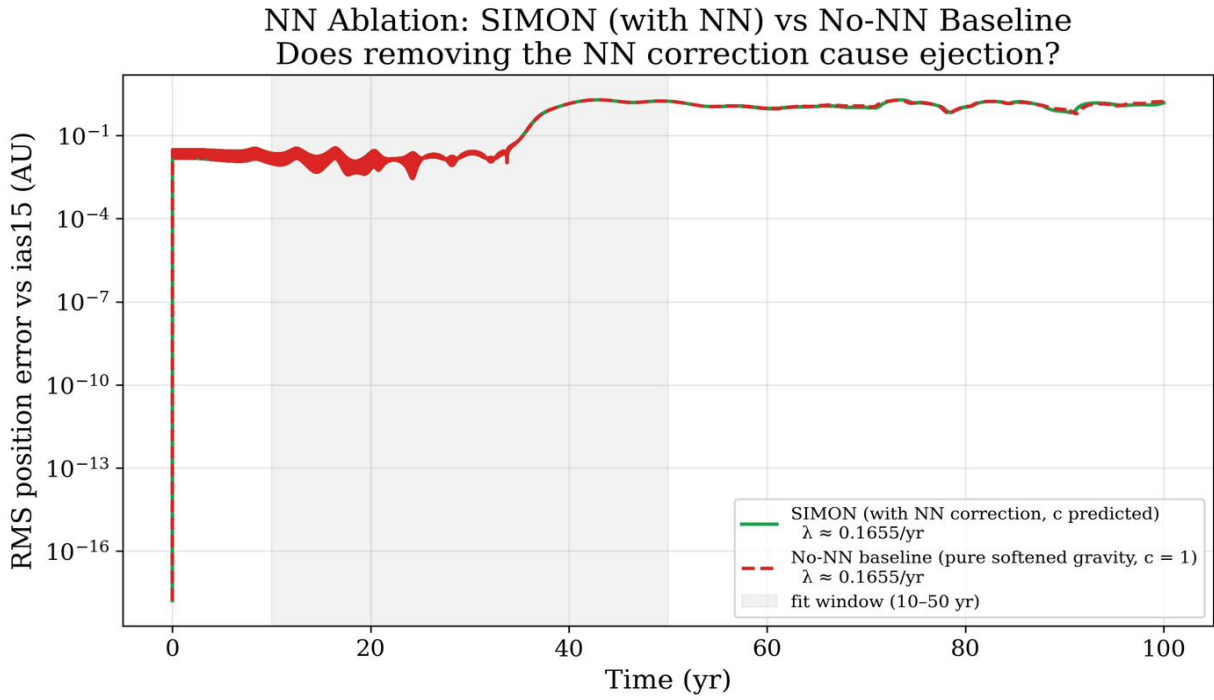


Figure 4. RMS position error vs ias15 for SIMON (green, Run D: adaptive ON, NN ON) and No-NN baseline (red dashed, Run C: adaptive ON, NN OFF). Both models remain bounded throughout the 100-year simulation. The 8.5% difference in final RMS (1.54 vs 1.67 AU) quantifies the NN correction's contribution: force accuracy improvement at close-encounter steps, not orbital stability. This figure distinguishes Run C from Run B: removing analytic direction (Run B) causes ejection; removing the NN correction (Run C) does not.

5. Paper-Ready Language and Claims

This section provides the exact framing to use in the V3 paper, incorporating all mentor feedback and the decisions made in our discussion.

5.1 Correct Scoping of the n=1 Result

Do not use: 'The Vector NN causes body ejection.' (This implies a universal causal claim from a single trajectory.) **Use instead:** 'For the tested initial condition, the Vector NN architecture produces unbounded trajectories. The gate-sensitivity analysis demonstrates this arises from the accumulation of non-radial force components at close encounters, not from a single anomalous prediction: tightening the gate threshold to below the NN's own demonstrated accuracy ($\cos < 0.9999985$) produces identical ejection timing and identical final RMS, confirming the result is insensitive to the specific gate parameter chosen.'

5.2 Correct Role of the NN Correction (replaces Section 7.3a)

Do not use the Section 7.3a argument that SIMON is equivalent to a Magnitude NN because c is approximately 1 for 99% of steps. This claim directly contradicts the ablation result that removing the NN costs 8.5% accuracy. **Use instead:** 'The NN correction contributes to force accuracy at close encounters. An ablation removing the NN correction while retaining adaptive stepping and analytic force direction yields bounded orbits with 8.5% larger final RMS error (1.67 vs 1.54 AU, Experiment 6). The NN is not the primary stability mechanism - that role belongs to adaptive sub-stepping - but it provides a measurable and quantified accuracy improvement at the 1.1% of steps involving close encounters.'

5.3 Correct Structure for the Paper's Architecture Comparison Section

The V3 paper section should follow this order:

- **First - Gate sensitivity analysis (Table 4):** This is the scientific result. Present the mechanistic argument (0.04 deg x 2,500 steps = approx 100 degrees accumulated non-radial impulse) and the gate threshold insensitivity as the primary evidence.
- **Second - Quantitative summary table (Table 5):** Present the lambda values, final RMS, and bounded/ejection status. Use scoped language referencing the tested initial condition.
- **Third - Trajectory and divergence figures (Figures 2 and 3):** Present as illustrations of the mechanistic finding. Caption them explicitly as illustrations, not as standalone evidence.
- **Fourth - Four-run framework (Table 6):** Place this experiment as Run B in the ablation table, connecting it to Experiments 4 and 6.

5.4 Literature Connection

SIMON's analytic direction constraint connects to the equivariant neural network literature (NequIP, SchNet, DimeNet) from molecular dynamics, where rotational symmetry is enforced through architectural design rather than learned. SIMON achieves rotational equivariance for the force direction through physics-based computation, not through the NN. The Vector NN experiment demonstrates that this structural constraint is

not merely a design preference but a functional requirement for chaotic long-horizon stability.

6. Summary of What This Experiment Proves

Scientific finding: Analytic force direction enforcement is necessary, not merely sufficient, for long-horizon orbital stability in chaotic three-body neural integration. For the tested initial condition, replacing analytic direction with a learned direction of 0.04-degree accuracy causes body ejection. Gate-sensitivity analysis confirms this arises from the accumulation of non-radial force components over approximately 2,500 integration steps - a class of non-physical impulse that cannot be prevented by any gate-based quality filter. **Role in V3:** This experiment is Run B in the four-run ablation framework. It provides the experimental evidence for the second of V3's two necessary physical constraints. The NN correction's role (8.5% accuracy improvement, not stability) is established separately by Run C in Experiment 6.

End of Experiment 1 Paper-Ready Section - Architecture Comparison - SIMON V3 - May 2026